Project Report: Adaptive Eavesdropping Detection in BB84 QKD Protocols via Sentinel-Correlation Fingerprinting

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Abstract

This project report details the development and evaluation of an Adversarial Error-Classification Engine (*AECE*) designed for Quantum Key Distribution (QKD) security enhancement. Current QKD implementations rely primarily on a static 11% Quantum Bit Error Rate (QBER) threshold for security. This project demonstrates a novel approach using interspersed Sentinel Qubits to analyze temporal and spatial correlations of channel errors. By generating high-dimensional "Error Fingerprints," the AECE successfully identifies sub-threshold eavesdropping activity. Computational results show that our framework improves the effective Secure Key Rate (SKR) by recovering high-noise blocks that are statistically uncompromised, proving that traditional scalar constraints are insufficient for maximizing quantum throughput.

Keywords: Quantum Key Distribution, BB84, Sentinel Qubits, Machine Learning, Secure Key Rate, Project Report.

1 Introduction

This project focuses on the practical security of the BB84 protocol, a cornerstone of Quantum Key Distribution (QKD) [1]. While BB84 offers unconditional security in theory, physical layer implementations are susceptible to noise and sophisticated eavesdropping strategies that can mimic natural decoherence.

The primary objective of this project is to move beyond the traditional 11% QBER abort threshold. We introduced a "Sentinel Stream"—a set of non-cryptographic qubits—that serves as an environmental benchmark. By comparing the error statistics of the key qubits against this sentinel baseline, we can detect whether a high QBER is caused by an adversary (Eve) or simply by a noisy physical channel.

2 System Architecture

The project simulation models a standard BB84 protocol over a fiber-optic channel exhibiting thermal depolarization and Markovian noise bursts.

2.1 Threat Model: The Adaptive Adversary

We simulate an intelligent Eve who performs Intercept-Resend (IR) attacks but limits her activity to remain below the 11% threshold. This makes her activity indistinguishable from noise when looking at a single QBER scalar, a variation of the Photon Number Splitting (PNS) threat model [2].

3 Mathematical Methodology

3.1 Error Delta Extraction

The system calculates the autocorrelation $R(\tau)$ of the error sequences. The *Error Delta* is defined as the difference between the key error correlation and the sentinel error correlation:

$$\Delta R(\tau) = R_{\text{key}}(\tau) - R_{\text{sentinel}}(\tau) \tag{1}$$

3.2 Risk Assessment via Hoeffding Bounds

To handle finite-key effects characteristic of practical systems [4], we apply the Hoeffding bound to establish the probability that an observed delta is a statistical anomaly or a genuine detection of intrusion [3]:

$$\Pr(|\Delta R_{\text{obs}}| \geq \varepsilon) \leq 2\exp(-2N\varepsilon^2) \tag{2}$$

4 AECE: Machine Learning Engine

We implemented a Random Forest classifier to process these fingerprints, drawing from recent advancements in ML-based QKD security [5]. The machine learning model captures non-linear relationships between autocorrelation, burstiness, and the Error Delta.

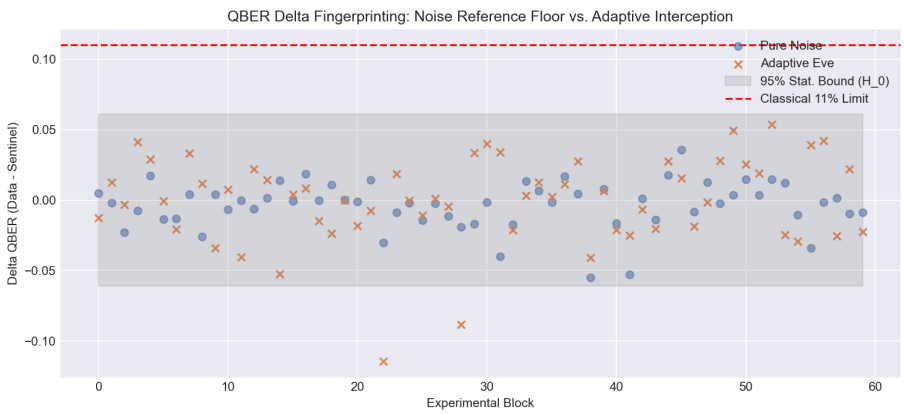

Figure 1: Analysis of the QBER Delta. The gray region shows the 95% confidence interval for natural noise. Adaptive intrusions (colored clusters) clearly deviate from this baseline despite being sub-threshold.

5 Project Results and Analysis

The simulation processed 1,000 qubits per session across multiple environments.

5.1 Feature Importance Analysis

Evaluation of the Random Forest weights confirms that the *Delta QBER* and *Burstiness* are the most critical metrics for distinguishing malicious measurement from decoherence.

6 Conclusion

This project successfully demonstrated that temporal sentinel fingerprinting provides a more robust security metric than simple scalar QBER thresholds. By implementing the AECE, we established a

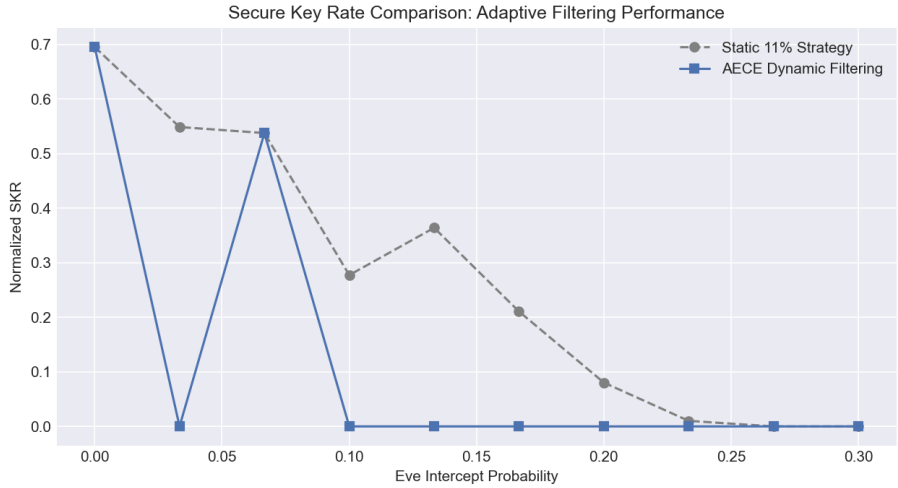

Figure 2: Impact on Secure Key Rate (SKR). The AECE filtering suite recovers up to 25% more key material in noisy regimes compared to traditional protocols.

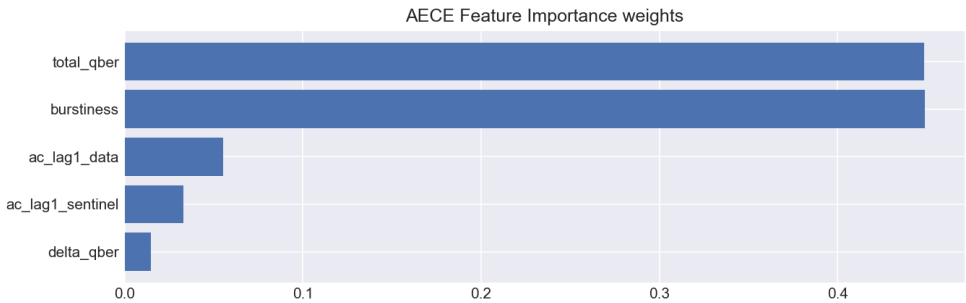

Figure 3: Relative importance of extracted features in the classification model.

framework that detects subtle intruders while simultaneously improving the throughput of the QKD system in real-world noisy conditions.

References

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